ENTHUSE, LEADER & ACHIEVER COURSE

PHASE : ALL ENTHUSIAST,MLU,MAZA,B,L,M,N,MAPX,MAPA,MAAX,MAAW,LAKSHYA

TARGET : PRE-MEDICAL 2026

Test Type : MAJOR

Test Pattern : NEET (UG)

TEST DATE : 05-04-2026

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	2	3	1	3	1	2	3	2	1	1	3	1	3	2	3	2	2	1	2	1	2	4	3	1	2	4	2	2	2	2
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	3	4	1	4	2	3	4	4	1	1	1	2	3	3	1	4	1	4	2	1	4	3	3	3	2	3	4	2	3	3
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	3	4	3	2	1	1	3	3	1	4	2	2	1	1	4	3	2	3	3	2	1	1	3	1	1	4	1	1	2	2
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	3	4	4	1	3	1	4	1	3	2	3	4	1	1	3	2	3	2	1	2	3	4	3	4	3	1	4	1	4	3
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	4	4	1	1	4	4	1	3	3	1	2	2	3	3	4	3	4	1	4	2	4	1	3	3	1	2	2	4	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	2	2	1	4	1	1	3	4	3	2	3	2	3	4	3	3	1	4	2	3	1	4	2	1	4	3	2	1	2	3

HINT – SHEET

1. Ans (2)

$v^2 = t \Rightarrow v = t^{1/2} \Rightarrow a = \frac{dv}{dt} = \frac{1}{2}t^{-1/2}$

$\text{At } t = 4s,\ a = \frac{1}{2}(4)^{-1/2} = \frac{1}{4}m/s^2$

2. Ans (3)

$H = \frac{u^2\sin^2\theta}{2g}$

$\text{Given} = \frac{u_1^2\sin^2 45^\circ}{2g} = \frac{u_2^2\sin^2 60^\circ}{2g}$

$\frac{u_1^2}{u_2^2} = \frac{\sin^2 60^\circ}{\sin^2 45^\circ} \Rightarrow \frac{u_1}{u_2} = \frac{\frac{\sqrt{3}}{2}}{\frac{1}{\sqrt{2}}} = \sqrt{\frac{3}{2}}$

3. Ans (1)

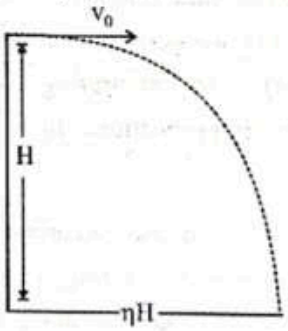
$v = kx$

$a = v\frac{dv}{dx} = (kx)(k)$

$a \propto x$

$a \propto \frac{v}{k} \Rightarrow a \propto v$

4. Ans (3)



$x = \eta H$

$v_0 T = \eta H$

$v_0\sqrt{\frac{2H}{g}} = \eta H$

$v_0 = \eta\sqrt{\frac{gH}{2}}$

$v = \sqrt{v_0^2 + 2gH} = \sqrt{\eta^2\frac{gH}{2} + 2gH}$

$v = \sqrt{\left(\frac{\eta^2+4}{2}\right)gH}$

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5. **Ans (1)**

From screw guage

zero error = + 0.20 mm

Diameter of Chalk = 3.5 – 0.2

= 3.3 mm

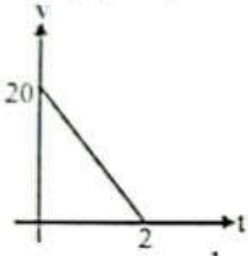
for vernier calliper

zero error = – 0.05 cm

Thus it must show = 0.33 – 0.05

= 0.28 cm

6. **Ans (2)**



Initial KE $= \frac{1}{2}mu^2$

$= \frac{1}{2} \times 2 \times (20)^2 = 400$

Final KE = 0

$\therefore$ Work done = – 400 J

7. **Ans (3)**

If power is constant then by using the relation P = F.v we can state that a.v = constant then using dv/dt = k/v we get $v = (2kt)^{1/2}$ then using $dx/dt = (2kt)^{1/2}$ on integrating we get displacement is directly proportional to $t^{3/2}$.

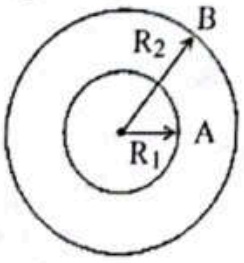
8. **Ans (2)**

Since ω is constant, v would also be constant. So, no net force or torque is acting on ring. The force experienced by any particle is only along radial direction, or we can say the centripetal force.

The force experienced by inner part, $F_1 = m\omega^2 R_1$

The force experienced by outer part, $F_2 = m\omega^2 R_2$

$\frac{F_1}{F_2} = \frac{R_1}{R_2}$



9. **Ans (1)**

Retardation in upward motion = $g(\sin\theta + \mu\cos\theta)$

$\therefore$ Force required just to move up $F_{up} = mg(\sin\theta + \mu\cos\theta)$

Similarly for down ward motion $a = g(\sin\theta - \mu\cos\theta)$

$\therefore$ Force required just to prevent the body sliding down F_{dn}

$= mg(\sin\theta - \mu\cos\theta)$

According to problem $F_{up} = 2F_{dn}$

$\Rightarrow mg(\sin\theta + \mu\cos\theta) = 2mg(\sin\theta - \mu\cos\theta)$

$\Rightarrow \sin\theta + \mu\cos\theta = 2\sin\theta - 2\mu\cos\theta$

$\Rightarrow 3\mu\cos\theta = \sin\theta$

$\Rightarrow \tan\theta = 3\mu$

$\Rightarrow \theta = \tan^{-1}(3\mu) = \tan^{-1}(3 \times 0.25) = \tan^{-1}(0.75)$

$= 37°$

10. **Ans (1)**

The coin falls behind him it means the velocity of train was increasing otherwise the coin fall directly into the hands of thrower.

11. **Ans (3)**

$\because$ Density of cone and cylinder is same

$M_{cylinder} = M_1 = \rho\pi R^2 H$

$M_{cone} = M_2 = \frac{\rho\pi R^2 H}{3} = \frac{M_1}{3}$

$y_{COM} = \frac{M_1 y_1 + M_2 y_2}{M_1 + M_2}$

$= \frac{M_1(H/2) + M_2(H + H/4)}{M_1 + M_2}$

$y_{COM} = \frac{11H}{16} < H$

12. **Ans (1)**

For perfect elastic e = 1

For perfect inelastic e = 0

For inelastic 0 < e < 1

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13. Ans (3)

Take same angular speed; linear speed $v \alpha r$, Use E

$= \frac{1}{2}mv^2$, $F = \frac{mv^2}{r}$, $P = mv$.

$\omega_A = \omega_B$

$\frac{V_A}{V_B} = \frac{\omega_A r_A}{\omega_B r_B} = \frac{3\ell}{\ell} = 3 \rightarrow R$

$\frac{E_A}{E_B} = \frac{\frac{1}{2} \times 3mV_A^2}{\frac{1}{2} \times 2mV_B^2} = \frac{3}{2} \times (3)^2 = \frac{27}{2} \rightarrow P$

$\frac{F_A}{F_B} = \frac{3m\omega_A^2 r_A}{2m\omega_B^2 r_B} = \left(\frac{3}{2} \times \frac{1}{1} \times \frac{3}{1}\right) = \frac{9}{2} \rightarrow S$

$\left(\frac{P_A}{P_B}\right)^2 = \left(\frac{3m\,V_A}{2m\,V_B}\right)^2 = \left(\frac{3}{2} \times 3\right)^2 = \frac{81}{4} \rightarrow Q$

14. Ans (2)

Tangential acceleration from $v = \frac{dx}{dt}$ and centripetal acceleration $a_c = \frac{v^2}{r}$,

$x = 2t^2 \Rightarrow v = \frac{dx}{dt} = 4t$

At $t = 2$:

$v = 8 m/s$

Tangential acceleration:

$a_t = \frac{dv}{dt} = 4$

Centripetal acceleration:

$a_c = \frac{v^2}{r} = \frac{64}{8} = 8$

Total acceleration:

$a = \sqrt{a_t^2 + a_c^2} = \sqrt{16 + 64} = \sqrt{80} = 4\sqrt{5}$

15. Ans (3)

For polytropic process PV^m = constant

$C = C_v + \left(\frac{R}{1-m}\right)$ (Here $m = \frac{3}{2}$ and $C_v = \frac{3R}{2}$)

$\Rightarrow C = \frac{3R}{2} + \frac{R}{1-\frac{3}{2}} = \frac{3R}{2} - 2R = -0.5R$

$Q = nC\Delta T = 1\,(-0.5\,R)\,(-26) = +13R$

16. Ans (2)

$Q_{abs} = W$ = Area of curve

$= \pi[a]\,[b]$

$= \pi[10 \times 10^{+3}] \times [10 \times 10^{-3}]$

$= 100\,\pi\,J$

17. Ans (2)

$P \times t \times \eta = ms\Delta T$

$2000 \times t \times 0.7 = 2 \times 1000 \times 4.2 \times 50$

$t = 300\ s$

18. Ans (1)

$\frac{\rho_1}{\rho_2} = \frac{3}{4}, \frac{s_1}{s_2} = \frac{4}{3}$

$\frac{H_1}{H_2} = \frac{\rho_1}{\rho_2} \times \frac{s_1}{s_2} = \frac{3}{4} \times \frac{4}{3} = 1$

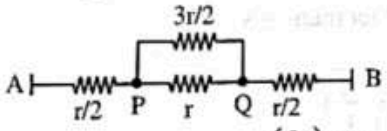
19. Ans (2)

$\because R = \frac{L}{KA} = r[R \propto L]$

$R_{AP} = \frac{r}{2}$; $R_{PQ\,(Straight)} = r$

$R_{PQ\,(Bend)} = \frac{3r}{2} = 1.5r$

$R_{QB} = \frac{r}{2}$



$R_{eq(PQ)} = \frac{R_1R_2}{R_1 + R_2} = \frac{\left(\frac{3r}{2}\right)(r)}{\frac{3r}{2} + r} = \frac{3r}{5}$

$R_{eq(AB)} = \frac{r}{2} + \frac{3r}{5} + \frac{r}{2} = 1.6r$

$\because \Delta T \propto R$ [V = IR]

$\frac{\Delta T_{(PQ)}}{\Delta T_{(AB)}} = \frac{R_{PQ}}{R_{AB}} = \frac{(3r/5)}{(1.6r)} = \frac{3}{8}$

$\Delta T_{PQ} = \frac{3}{8}\Delta T_{AB} = \frac{3}{2} \times 120$

$\Delta T_{PQ} = 45°C$

20. Ans (1)

The power radiated by a filament is $P = e\sigma AT^4$

where e = emissivity, σ = Stefan's constant,

T= surface temperature. or $e_A T_A^4 = e_B T_B^4$

$\Rightarrow \frac{T_A}{T_B} = \left(\frac{e_B}{e_A}\right)^{1/4}$

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21. Ans (2)

$f_1 = \frac{1}{2\pi}\sqrt{\frac{k_1}{m}}$ $\quad f_2 = \frac{1}{2\pi}\sqrt{\frac{k_2}{m}}$

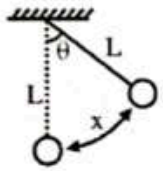
$f = \frac{1}{2\pi}\sqrt{\frac{k_1 + k_2}{m}}$

Solving we get $f = \sqrt{f_1^2 + f_2^2}$

22. Ans (4)

As pendulum starts from EP

So equation of SHM is



$\theta = \theta_0 \cos \omega t$ where $\omega = \frac{2\pi}{T}$

$\omega = \frac{2\pi}{2\pi\sqrt{\frac{L}{g}}} = \sqrt{\frac{g}{L}} \Rightarrow \theta = \theta_0 \cos\sqrt{\frac{g}{L}}t$

Linear displacement = $x = L\theta$

$x = L\theta_0 \cos\sqrt{\frac{g}{L}}t$

23. Ans (3)

$T = 2(0.5 + 1.5)$

$T = 4$ second, $\omega = \frac{2\pi}{4} = \frac{\pi}{2}$

Start M.P.

$x = A \sin \omega t$

$v = A\omega \cos \omega t$

$t = 0.5, v = 3$

$3 = \frac{A\pi}{2}\cos\left(\frac{\pi}{2} \times 0.5\right)$

$A = \frac{6\sqrt{2}}{\pi}$

M.P.

$V = A\omega$

$V = \frac{6\sqrt{2}}{\pi} \times \frac{\pi}{2} = 3\sqrt{2}$ m/s

24. Ans (1)

$\frac{2\pi}{\lambda} = \frac{\pi}{4} \quad \lambda = 8$

$N - N = \frac{\lambda}{2} = \frac{8}{2} = 4m$

25. Ans (2)

For $f_1 = 504$ Hz, third harmonic is $3 \times 504 = 1512$ Hz

For $f_2 = 496$Hz, third harmonic is $3 \times 496 = 1488$ Hz

Case-I : $(1512 - 1510) = 2$ b/s

Case-II : $(1488 - 1510) = 22$ b/s

So $f_1 = 504$ Hz

26. Ans (4)

$I_R = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$

Here, I_1=I, I_2=9I, and the resultant intensity I_R=7I.

Let's plug these values into the formula:

$7I = I + 9I + 2\sqrt{I \cdot 9I}\cos\phi$
$7I = 10I + 2\sqrt{9I^2}\cos\phi$
$7I = 10I + 2(3I)\cos\phi$
$7I = 10I + 6I\cos\phi$

Now, solve for cosϕ: $-3I=6I\cos\phi$

$\cos\phi = -\frac{3I}{6I} = -\frac{1}{2}$

The angle ϕ for which $\cos\phi=-1/2$ is **120°**.

27. Ans (2)

Sound wave ⟶ Longitudinal wave

Strings ⟶ Transverse wave

Long spring ⟶ Mixed

Light ⟶ EMW

28. Ans (2)

Heat lost = Heat gain

$ms_1 (32 - 20) = ms_2 (40 - 32)$

$12s_1 = 8s_2$

$\frac{s_1}{s_2} = \frac{2}{3}$

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29. Ans (2)

Let $[G] \propto c^x g^y p^z$

by substituting the following dimensions

$[G] = [M^{-1}L^3T^{-2}]$, $[c] = [LT^{-1}]$, $[g] = [LT^{-2}]$

$[p] = [ML^{-1}T^{-2}]$

and by comparing the powers of both sides

we can get $x = 0, y = 2, z = -1$

$\therefore [G] \propto c^0 g^2 p^{-1}$

30. Ans (2)

$$n_1u_1 = n_2u_2 \Rightarrow n_2 = n_1\frac{u_1}{u_2}$$

$$n_2 = 5\left[\frac{5kg}{\alpha kg}\right]\left[\frac{m}{\beta m}\right]^2\left[\frac{sec}{\gamma\, sec}\right]^{-2}$$

$$J \longrightarrow \frac{kg \times m^2}{sec^2}$$

31. Ans (3)

to lift

$T = 2g$

$T - 1g = 1 \times a_1$

$\boxed{a_1 = g}$

After monkey stop

$$a_2 = \frac{(2-1)g}{2+1} = \frac{g}{3}$$

$$\Delta a = g - \frac{g}{3} = \frac{2g}{3}$$

32. Ans (4)

$$s = \frac{t^2}{3} \Rightarrow \frac{ds}{dt} = \frac{2t}{3}$$

i.e. $v = \frac{2t}{3}$

work done = ΔKE

$$WD = \frac{1}{2}m\left(v_f^2 - v_i^2\right)$$

at $t = 0$ $v_i = 0$

at $t = 2$ sec $v_f = \frac{4}{3}$ m/s

$$WD = \frac{1}{2} \times 3\left(\frac{16}{9} - 0\right)$$

$$= \frac{8}{3}J$$

33. Ans (1)

MI of given ring = $\frac{MR^2}{2}$

Where $\pi R = L \Rightarrow R = \frac{L}{\pi}$

So, $I = \frac{ML^2}{2\pi^2}$

34. Ans (4)

Couple is a pair of two equal and opposite force having different lines of action.

So $F_{net} = 0$

But $\tau_{net} \neq 0$

35. Ans (2)

To keep the disc horizontal

Net torque = 0 (about point A)

$Mg(R) - T(2R) = 0$

$MgR - mg(2R) = 0$ $(T = mg)$

$\boxed{m = \frac{M}{2}}$

36. Ans (3)

$$g = \frac{GM}{R^2}$$

$$\therefore \quad M = r \times \frac{4}{3}\pi R^3 \Rightarrow g = \frac{G \times 4\pi R^3 \times \rho}{3 \times R^2}$$

$$\Rightarrow \rho = \frac{3g}{4\pi GR}$$

37. Ans (4)

$$U = -\frac{GMm}{R}$$

Escape velocity $(v) = \sqrt{\frac{2GM}{R}}$

$$+54 = +\frac{GM}{R}(3)$$

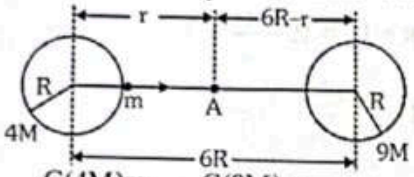
$$\frac{GM}{R} = \frac{54}{3} = 18$$

$$v = \sqrt{2 \times 18} = 6 \text{ m/s}$$

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38. Ans (4)

The gravitational force on projectile of mass m at a distance r from a planet of mass 4M is zero



$$\therefore \frac{G(4M)m}{r^2} = \frac{G(9M)m}{(6R-r)^2}$$

$$\Rightarrow 4(6R-r)^2 = 9r^2$$

$$\Rightarrow 2(6R-r) = 3r$$

$$\Rightarrow r = \frac{12}{5}R \Rightarrow r = 2.4\ R.$$

39. Ans (1)

$$W = U = \frac{1}{2}\frac{YA}{L}(\Delta L)^2$$

$$= \frac{1}{2} \times 2 \times 10^{11} \times 1 \times 10^{-6} \times (10^{-3})^2$$

$$W = 0.1\ J$$

40. Ans (1)

For equilibrium

$$8\ cm \times \rho_y \times g + 2\ cm \times \rho_{Hg} \times g = 10\ cm \times \rho_x \times g$$

$$8 \times \rho_y + 2 \times 13.6 = 10 \times 3.36$$

$$\rho_y = \frac{33.6 - 27.2}{8} = 0.8\ g/cm^3$$

41. Ans (1)

$$v = \sqrt{2g(H-h)}$$

42. Ans (2)

$$F = \frac{YA}{\ell} \times \Delta\ell$$

$$\text{slope (m)} = \tan\theta = \frac{YA}{\ell}$$

$$\tan\theta \propto A$$

43. Ans (3)

Surface tension of water is more than surface tension of soap solution.

44. Ans (3)

$$0.1\ mm = \frac{10-9}{10}(MSD)$$

$$MSD = 1mm$$

45. Ans (1)

$$\lambda = 2[0.49 - 0.16]$$

$$\lambda = 2[0.33]$$

$$n = \frac{v}{\lambda} = \frac{330}{2 \times 0.33} = 500\,Hz$$

46. Ans (4)

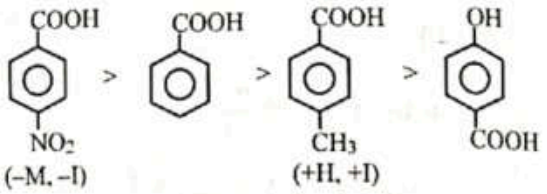
Solution:

Acidic strength (K_a) $\propto$ Stability of conjugate base.

$$\propto \frac{-M,\ -I}{+M,\ +I}$$

$\propto$ Delocalisation of charge in conjugate base.

Overall acidic strength order:

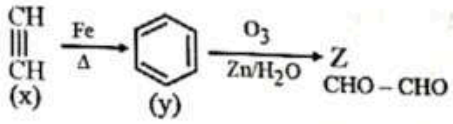


Final Answer: (4)

47. Ans (1)

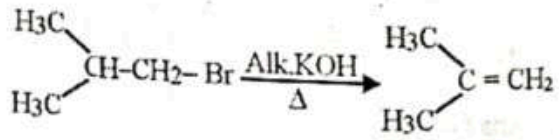
For testing of halogens, Nitric acid is added to the sodium extract because if CN^- or S^{2-} are present then they will be oxidised and removed before the test of halides.

48. Ans (4)



49. Ans (2)

Solution:



Only 1 alkene

Final Answer: Option (2)

50. Ans (1)

Cyclic, planar, completely conjugated system follow Huckel rule $(4n+2)\ \pi\ e^-$ rule) are aromatic

51. Ans (4)

Gauche form is less stable than antiform because of more steric strain in gauche.

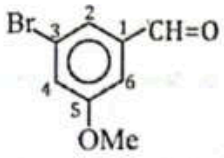
52. Ans (3)

neutral R.S. is more stable.

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53. **Ans (3)**

Solution:

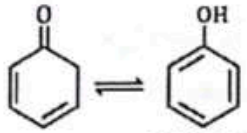


3-Bromo-5-methoxybenzaldehyde

Final Answer: (3)

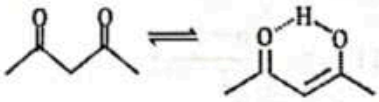
54. **Ans (3)**

Solution:

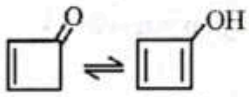


(more enol content)

Stable enol

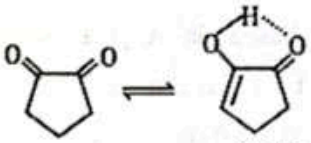


Stable enol (more enol content)



Keto content Maximum

Anti Aromatic (unstable)



(more enol content)

Stable enol

Final Answer

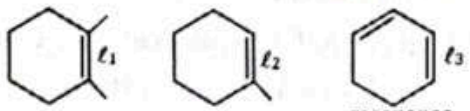
The correct option is (3)

55. **Ans (2)**

Solution:

In alkenes C=C bond length increases due to hyperconjugation effect and resonance.

Bond length ∝ reso ∝ H-effect



Hyper conjugable Hydrogen or αH=10

αH=7

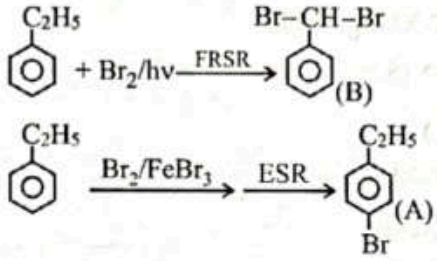
resonance $\therefore \ell_3 \uparrow$

so bond length order is : $\ell_3 > \ell_1 > \ell_2$.

Final Answer:

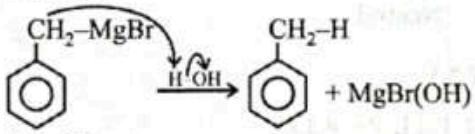
The correct option is (2)

56. **Ans (3)**



57. **Ans (4)**

Concept:



Ans.(4)

58. **Ans (2)**

$\omega = 0.25$ gm

$V_1 = 40$ mL

$T_1 = 300$ K

$P_1 = (725 - 25) = 700$ mm

N.T.P.

$T_1 = 273$ K

$P_2 = 760$ mm

$V_2 = ?$

$$V_2 = \frac{P_1V_1}{T_1} \times \frac{T_2}{P_2}$$

$$= \frac{700 \times 40}{300} \times \frac{273}{760}$$

$= 33.52$ ml

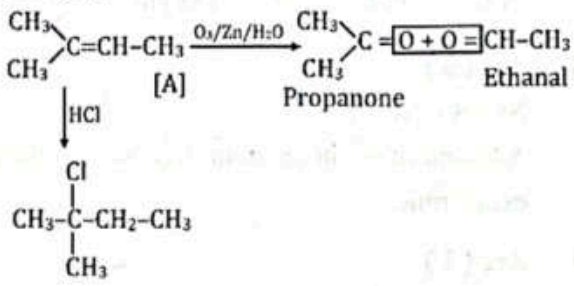
$$\% N = \frac{\text{Vol. at N.T.P.}}{8 \times \omega}$$

$$= \frac{33.52}{8 \times 0.25} = 16.76\%$$

59. **Ans (3)**

NCERT Fact

60. **Ans (3)**



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61. Ans (3)

NCERT-XI, Pg. # 90

$\Delta Hc_g \Rightarrow [S > Se > Te > Po > O]$

62. Ans (4)

NCERT-XI, Pg. # 94

Cl_2O_7 Acidic

Na_2O Basic

As_2O_3 Amphoteric

N_2O Neutral

63. Ans (3)

NCERT-11, Pg. # 130

C_2 $(12e^-)$ = $\sigma 1s^2$, $\sigma^* 1s^2$, $\sigma 2s^2$, $\sigma^* 2s^2$, $[\pi(2p_x^2) = \pi(2P_y)^2]$

BO = 2[Both are pure π bond]

Because of the presence of four electrons in two pi molecular orbital

64. Ans (2)

Due to presence of Inter molecular H-bond.

65. Ans (1)

NCERT-XI, Pg. # 116

$XeF_4 \Rightarrow sp^3d^2$ [bp = 4, $\ell p = 2$]

$BrF_5 \Rightarrow sp^3d^2$ [bp = 5, $\ell p = 1$]

$SF_4 \Rightarrow sp^3d$ [bp = 4, $\ell p = 1$]

66. Ans (1)

NCERT-11, Chemical bonding M.O.T Postulates

67. Ans (3)

NCERT-XI, Pg. # 108

C = O < N = O < C = C < C = N

121 122 133 138 pm

68. Ans (3)

NCERT-XI, Pg. # 91

Addition of e^- in an atom may be endothermic or exothermic.

69. Ans (1)

NCERT-XI, Pg. # 120

70. Ans (4)

$\boxed{LE \propto |q_1 q_2|}$

[Size of Anion ↑ ⇒ Covalent character ↑]

[BP ∝ molecular weight ∝ Vander waal force of Attraction]

Bond strength ⇒ F–F < Cl–Cl > Br–Br

71. Ans (2)

NCERT-XI, Pg. # 80

Z = 106 (Sg)

Z = 113 (Nh)

Z = 108 (Hs)

Z = 117 (Ts)

72. Ans (2)

Total spectral line = $\frac{n(n-1)}{2} = 15$

$\therefore n^2 - n = 30$

or n = 6

$\therefore$ lines in visible region = $n - 2 = 4$

73. Ans (1)

NCERT, Pg. # 211

74. Ans (1)

This question is based on Angular nodes (ℓ), Radial nodes (n - ℓ - 1) Total nodes (n - 1)

(a) 5s (n = 5, ℓ = 0) has 0 angular nodes (ℓ = 0) This matches (Q).

(b) 3dyz (n = 3, ℓ = 2) has O radial nodes (n - ℓ - 1 = 3 - 2 - 1 = 0) This matches (S).

(c) 4dxy (n = 4, ℓ = 2) has 1 radial node (n - ℓ - 1 = 4 - 2 - 1 = 1) and a total of 3 nodes (n - 1 = 4 - 1 = 3) . This matches (P) and (R).

(d) 3p (n = 3, ℓ = 1) has 1 radial node (n - ℓ - 1 = 3 - 1 - 1 = 1) This matches (P).

Therefore, the correct pairings are

(a) ⟶ (Q), (b)⟶(S), (c) ⟶ (P,R), and (d) ⟶ (P).

Final Answer: (1)

75. Ans (4)

NCERT, Pg.# 56

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76. **Ans (3)**

$PtCl_4 + 5XeF_2 \longrightarrow PtF_6 + 4ClF + 5Xe$

77. **Ans (2)**

MnO_4^-, $CrO_2Cl_2 \longrightarrow +7, +6$

MnO_2, $FeCl_3 \longrightarrow +4, +3$

$MnCl_2$, $CrCl_3 \longrightarrow +2, +3$

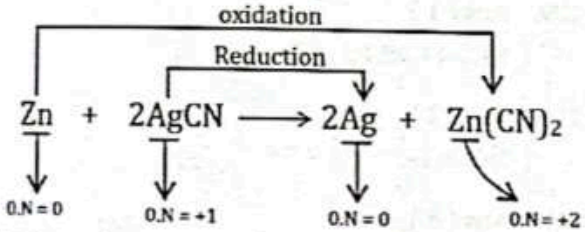
$[NiCl_4]^{2-}$, $Hg_2Cl_2 \longrightarrow +2, +1$

78. **Ans (3)**

Question Explanation : The question asks which reaction is redox reaction.

Concept : This question is based on Redox Reaction.

Solution :



This reaction is Redox Reaction

– Other reaction involve double displacement, decomposition or acid-base reactions-without oxidation state changes.

Final Answer: option (3)

79. **Ans (3)**

$$2NO_2 + \frac{1}{2}O_2 \rightleftharpoons N_2O_5 \; ; \; k_p$$

So $4NO_2 + O_2 \rightleftharpoons 2N_2O_5 \; ; \; k_p^2$

So $2N_2O_5 \rightleftharpoons 4NO_2 + O_2; \; \frac{1}{k_p^2}$

80. **Ans (2)**

$2CO_2(g) \rightleftharpoons 2CO(g) + O_2(g)$ V = 5L

Initially	2	0	0
at Eq	2 – 1.2 = 0.8	1.2	0.6

$n_{eq} = 0.8 + 1.2 + 0.6$

$= 2.6$

81. **Ans (1)**

	A	+ B	⇌ C	+ D
initial	1	1	0	0
At eq^m	1 – x	1 – x	x	x

$$k_{eq} = \frac{x^2}{(1-x)^2} \Rightarrow \frac{x}{1-x} = \sqrt{k}$$

$$\Rightarrow x = \frac{\sqrt{k}}{1+\sqrt{k}}$$

Degree of dissociation $= \frac{x}{1} = x = \frac{\sqrt{k}}{1+\sqrt{k}}$

82. **Ans (1)**

H_2	+ I_2	⇌ 2HI
1	1	0
1–α	1–α	2α
= 1–0.8	1–0.8	2 × 0.8
= 0.2	= 0.2	= 1.6

$$\therefore K_C = \frac{1.6 \times 1.6}{0.2 \times 0.2} = 64$$

83. **Ans (3)**

Catalyst do not change position of equilibrium.

84. **Ans (1)**

$3Fe_{(s)} + 4H_2O(\ell) \longrightarrow Fe_3O_4(s) + 4H_2 \uparrow$

$3 \times 56g \quad 4 \times 18\,g \quad 232\,gm$

assume 1mL water = 1g

L.R is Fe

$$\therefore 56\,g\,Fe \text{ gives} = \frac{232}{3}\,g\,Fe_3O_4 = 77.33\,g$$

85. **Ans (1)**

$$\% \text{ oxygen} = \frac{16}{40} \times 100 = 40\%$$

86. **Ans (4)**

(a) Liquid to gas ⟶ Entropy increases

(b) $\underset{\oplus}{\Delta G} = \underset{\oplus}{\Delta H} - \underline{T\Delta G}$ For ΔG to be +ve at all T, ΔS must be –ve

(c) Reversible process, $\Delta S = 0$,

87. **Ans (1)**

NCERT, Pg. # 55

88. **Ans (1)**

NCERT, Pg. # 185

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89. **Ans (2)**
Reaction (b) doesn't has one mole reactant in reaction (c) gaseous atoms are not formed.

90. **Ans (2)**
Moles of H_2SO_4 = 1 mol
g eq. of H_2SO_4 = moles × V.F. = 1 × 2 = 2 g eq.
ΔH due to 2 g eq. = $-13.7 \times 2 = -27.4$ kCal

91. **Ans (3)**
NCERT-XI, Pg. # 158

92. **Ans (4)**
NCERT XI Pg : 111-112

93. **Ans (4)**
NCERT-XI Pg. # 146

94. **Ans (1)**
Maltose = a-D glucose $(C_1) \longrightarrow (C_4)\alpha-D$ glucose

95. **Ans (3)**
NCERT, Pg. # 136

96. **Ans (1)**
NCERT, Pg. # 136

97. **Ans (4)**
NCERT-XI, Pg # 150

98. **Ans (1)**
NCERT-XI, Pg. # 137

99. **Ans (3)**
NCERT-XI, Pg. # 138, 139, 140, 142

100. **Ans (2)**
NCERT Pg.159

101. **Ans (3)**
NCERT Pg. # 160

102. **Ans (4)**
NCERT Pg. # 161-164

103. **Ans (1)**
NCERT-XI, Pg. # 156, 158

104. **Ans (1)**
NCERT-XI, Pg. # 157

105. **Ans (3)**
NCERT-XI, Pg. # 175, 176, 177, 178

106. **Ans (2)**
NCERT-XI, Pg # 248

107. **Ans (3)**
NCERT-XI, Pg. # 177
Explain Question: The question asks that Ethylene initiate flowering and synchronise fruit set in which of the following.
Concept: Ethylene.
Solution: Ethylene is used to initiate flowering and for synchronising fruit set in Pineapples. So, the correct answer is: 3.
Final answer: Option 3

108. **Ans (2)**
NCERT-XI, Pg. # 245, 247, 250

109. **Ans (1)**
NCERT XI Pg. # 242

110. **Ans (2)**
NCERT-XI, Pg. # 26

111. **Ans (3)**
NCERT-XI, Pg. # 28, Fig. 3.2(b)

112. **Ans (4)**
NCERT-XI, Pg. # 30, 32, 33, 34

113. **Ans (3)**
NCERT-XI, Pg. # 27

114. **Ans (4)**
NCERT-XI, Pg. No. # 30 & 32

115. **Ans (3)**
NCERT-XI, Pg.# 39

116. **Ans (1)**
NCERT-XI, Pg. # 11

117. **Ans (4)**
NCERT-XI, Pg. # 17, 18

118. **Ans (1)**
NCERT-XI, Pg. # 20

119. **Ans (4)**
NCERT-XI, Pg # 21, Fig. 2.4(a)

120. **Ans (3)**
NCERT-XI, Pg # 20, 21

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121. **Ans (4)**
NCERT-XI, Pg. # 15, 16

122. **Ans (4)**
NCERT-XI, Pg.# 4

123. **Ans (4)**
NCERT-XI, Pg. # 4, 5

124. **Ans (1)**
NCERT-XI Pg # 10, Fig. 1.1

125. **Ans (1)**
NCERT XI, Pg. No. # 95, 96

126. **Ans (4)**
Module Pg.# 148

127. **Ans (4)**
NCERT-XI, Pg. # 94

128. **Ans (1)**
NCERT-XI, Pg. # 100

129. **Ans (3)**
NCERT-XI, Pg. # 134,135

130. **Ans (3)**
NCERT-XI, Page No. # 129, 133, 137, 138

131. **Ans (1)**
NCERT-XI, Pg. # 121, 122

132. **Ans (2)**
NCERT XI Pg. No. # 127

133. **Ans (2)**
NCERT-XI, Pg. # 166

134. **Ans (3)**
NCERT-XI, Pg, # 123, 126

135. **Ans (3)**
NCERT-XI, Pg. # 123, 124, 126,127

136. **Ans (4)**
NCERT, Pg # 45

137. **Ans (3)**
NCERT, Pg. # 48

138. **Ans (4)**
NCERT, Pg. # 39

139. **Ans (1)**
NCERT, Pg. # 40, 41, 45, 49

140. **Ans (4)**
NCERT, Pg. 48

141. **Ans (2)**
NCERT, Pg. # 45

142. **Ans (4)**
Module

143. **Ans (1)**
Module

144. **Ans (3)**
Module

145. **Ans (3)**
NCERT-XI, Pg. # 113 (OLD)

146. **Ans (1)**
NCERT, Pg. # 83

147. **Ans (2)**
NCERT Pg. # 82,83,84

148. **Ans (2)**
NCERT, Pg. # 195

149. **Ans (4)**
NCERT, Pg. # 193, 194

150. **Ans (2)**
NCERT, Pg. # 202

151. **Ans (2)**
NCERT Pg. # 288

152. **Ans (2)**
NCERT, Pg. # 199, 200

153. **Ans (1)**
NCERT, Pg. # 191

154. **Ans (4)**
NCERT, Pg. # 190

155. **Ans (1)**
NCERT, Pg. # 187

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156. **Ans (1)**
NCERT Pg # 185 (E, H)

157. **Ans (3)**
NCERT Pg. # 183

158. **Ans (4)**
NCERT, Pg. # 209

159. **Ans (3)**
NCERT, Pg. # 212, 213

160. **Ans (2)**
NCERT, Pg # 205-206

161. **Ans (3)**
NCERT, Pg. # 291

162. **Ans (2)**
NCERT, Pg. # 212

163. **Ans (3)**
NCERT, Pg. # 213

164. **Ans (4)**
NCERT Pg. # 231.

165. **Ans (3)**
NCERT Pg. # 231, 232

166. **Ans (3)**
NCERT Pg # 234 (18.4.1)

167. **Ans (1)**
NCERT (18.3.1)

168. **Ans (4)**
NCERT XI Pg. # 247

169. **Ans (2)**
NCERT Pg.#246-247.

170. **Ans (3)**
NCERT Pg.#242-243.

171. **Ans (1)**
NCERT XI Pg. # 249, 250

172. **Ans (4)**
NCERT, Pg. # 243

173. **Ans (2)**
NCERT, Pg. # 245

174. **Ans (1)**
NCERT, Pg. # 243, 244

175. **Ans (4)**
NCERT, Pg. # 243

176. **Ans (3)**
NCERT, Pg. # 217, 218

177. **Ans (2)**
NCERT Pg. # 218

178. **Ans (1)**
NCERT Pg. No. 218

179. **Ans (2)**
NCERT Pg # 225

180. **Ans (3)**
NCERT Pg.#225

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